

# 00SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining**

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### COURSE OUTCOME COVERED

**CO3:** Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)  
Assessment Tool Weightage: 30%

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**QUESTIONS: 3**

**Total Marks:30**

Annexure C

Case study

| <i>Criteria</i>  | Understanding of Case (2.5) | Application of Concepts (2.5) | Analysis (2) | Solution Design (2) | Clarity (1) | <i>Total(10)</i> |
|------------------|-----------------------------|-------------------------------|--------------|---------------------|-------------|------------------|
| <i>Weightage</i> | 25%                         | 25%                           | 20%          | 20%                 | 10%         | <i>100%</i>      |
| <i>Q1</i>        |                             |                               |              |                     |             |                  |
| <i>Q2</i>        |                             |                               |              |                     |             |                  |
| <i>Q3</i>        |                             |                               |              |                     |             |                  |

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### ***Code of Conduct:***

*I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

### RUBRICS FOR CALCULATION:

| Criteria                | Weightage | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                     | Level 1 – Beginning                                   |
|-------------------------|-----------|------------------------------------------------------------------|------------------------------------------------|------------------------------------------|-------------------------------------------------------|
| Understanding of Case   | 25        | Thorough understanding ; clearly identifies all key issues       | Good understanding with most issues identified | Basic understanding ; some issues missed | Poor understanding ; problem not identified correctly |
| Application of Concepts | 25        | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts          | Fails to apply relevant concepts                      |
| Analysis                | 20        | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth        | Weak or no analysis; lacks reasoning                  |
| Solution Design         | 20        | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification  | Incorrect or no solution provided                     |
| Clarity                 | 10        | Clear, well-structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure            | Poor clarity; difficult to understand                 |

## ANNEXURE A

### Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

3. An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains “Offer” (Yes/No)
- Contains “Free” (Yes/No)
- Email Length (Short/Long)

### Sample Dataset

| Offer | Free | Length | Class    |
|-------|------|--------|----------|
| Yes   | Yes  | Short  | Spam     |
| Yes   | No   | Long   | Not Spam |
| No    | Yes  | Short  | Spam     |
| No    | No   | Long   | Not Spam |
| Yes   | Yes  | Long   | Spam     |
| No    | No   | Short  | Not Spam |

### Questions

1. Calculate prior probabilities:
  - $P(\text{Spam})$ ,  $P(\text{Not Spam})$
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
  - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes